Professional Issues

Supplementary document for the final-year project *Comparison of Machine Learning Algorithms*

Tayib Sheikh

BSc (Hons) Computer Science

Royal Holloway, University of London

2026

This document summarises the principal ethical, legal, and professional issues raised by the design, evaluation, implementation, and reporting of machine learning systems in the banking, medical, and cybersecurity case studies used in the project.

1. Purpose and scope

This project compared from-scratch implementations of k-Nearest Neighbours, Decision Trees, and Support Vector Machines on tabular classification tasks drawn from banking, medicine, and cryptocurrency security. Although the project was academic in purpose, the case studies themselves relate to decisions that can affect opportunity, safety, and investigation. For that reason, the work cannot be judged only as a technical benchmark. It must also be considered in terms of responsible design, legal compliance, data protection, transparency, and the wider consequences of how machine learning results are presented.

The central professional issue running through the project is that machine learning results can look impressive while still being unsafe or misleading if they are interpreted without sufficient care. A model may perform well numerically but still be unsuitable for real use if its evaluation is unrealistic, if its errors are operationally costly, or if the reporting conceals important caveats. In that sense, professional responsibility in machine learning begins not at deployment, but at system design, validation, and reporting.

2. Use of external software, datasets, and libraries

Development of the project depended on a mixture of original implementation and established external tools. The core learning logic for k-Nearest Neighbours, Decision Trees, and the Pegasos-based linear SVM was implemented from scratch. However, the wider experimentation framework relied on standard scientific Python tools and reference libraries, including NumPy, pandas, Matplotlib, and scikit-learn. Public benchmark datasets were also used, including Bank Marketing, Breast Cancer Wisconsin Diagnostic, Iris, and Bitcoin Heist.

From a professional perspective, the use of external software is entirely appropriate provided that it is acknowledged clearly and used within the conditions of the relevant licences. External tools can accelerate development and improve reproducibility, but they do not remove the responsibility to understand what the software is doing. In this project, scikit-learn was not used as a substitute for implementation, but as a reference point against which the custom models could be validated. That distinction matters because it affects both academic integrity and the credibility of the final results.

3. Licensing and attribution

Respecting intellectual property is a basic professional obligation in computing. The BCS Code of Conduct and the ACM Code of Ethics both place importance on honesty, competence, and proper acknowledgement of the work of others. In practical terms, that means software, datasets, figures, and ideas taken from external sources should not be presented as original work.

For this project, that principle applies in two ways. First, the datasets and external libraries used for benchmarking and validation must be cited properly in the report. Second, any baseline code, helper utilities, or visualisation approaches influenced by external examples must either be reimplemented independently or acknowledged openly. The project therefore distinguishes clearly between the from-scratch model implementations, which are the project's original technical contribution, and the external tools used to support experimentation, comparison, and reporting.

This distinction is especially important in machine learning projects because it is easy to blur the line between building a system and merely orchestrating other people's code. A professionally responsible report should make that boundary explicit. The strongest claim made in this project is not that every component was written without outside support, but that the core algorithms were implemented independently and then evaluated within a transparent framework.

4. Academic integrity and use of generative AI

Academic integrity requires that the final submission accurately reflects the student's own understanding and work. In computing, this includes both code and written analysis. Using external material without acknowledgement is plagiarism, but there is also a broader professional issue: presenting work that one does not properly understand can create a false impression of competence.

This issue has become more important with the rise of generative AI tools. In this project, generative AI was used only in a limited assistive capacity, for example to help with drafting, restructuring, wording refinement, and discussion support. However, the technical design, implementation decisions, debugging, evaluation logic, interpretation of results, and final judgement remained my own responsibility. Any AI assistance therefore functioned as a support tool rather than as a substitute for original academic work.

That distinction is professionally important. Generative AI can be useful for improving clarity or accelerating routine drafting, but it can also create risks if its output is copied uncritically, if it introduces unsupported claims, or if it obscures who is responsible for the final content. In a project centred on evaluation integrity, it would be contradictory to use AI in a way that undermines the integrity of the report itself. For that reason, any AI-supported material had to be checked against the actual repository evidence, revised where necessary, and incorporated only where it genuinely reflected the work carried out in the project.

A useful professional principle is that assistive tools may support the work, but they do not own the judgement. The responsibility for correctness, honesty, and academic defensibility remains with the author.

5. Data protection, privacy, and governance

All three project domains raise data governance issues. Banking datasets involve behavioural and financial attributes. Medical-style datasets involve diagnostic information. Blockchain-derived data may be publicly obtainable, but analysis of it can still contribute to surveillance, attribution, or reputational harm if handled irresponsibly.

The UK GDPR and related Information Commissioner's Office guidance make clear that personal data should be processed lawfully, proportionately, and transparently. Even in a student project using offline datasets, those principles still matter because they shape what responsible machine learning practice should look like. In particular, developers should think carefully about what data is used, why it is used, whether it is necessary, and what consequences may follow from the predictions made.

This project did not operationalise a live decision system, which reduces the immediate privacy risk. Even so, professional practice would require careful consideration of access control, explainability, auditability, retention, and data minimisation before any real deployment could be justified. A technically capable system can still be professionally deficient if it lacks a defensible governance framework.

6. Domain-specific professional issues

6.1 Banking and marketing prediction

The Bank Marketing dataset represents a commercial decision-support task: predicting whether a customer will subscribe to a term deposit. Although this is lower risk than clinical diagnosis or cybercrime detection, the professional issues are still significant. False positives waste outreach resources, while false negatives may cause the business to miss genuine opportunities. More importantly, if such a model were used operationally, it could shape who receives offers, contact, or attention.

This raises questions of fairness, transparency, and business accountability. Even when protected characteristics are not used directly, proxy variables may still produce skewed outcomes. A professional evaluation should therefore look beyond raw predictive success and consider stability, interpretability, and the commercial consequences of errors.

There is also a realism issue in the project's current benchmark. The authoritative Bank Marketing pipeline retains the feature `duration`, which is highly predictive but only fully known once a call has already taken place. That makes it useful for benchmarking, but much less suitable for a genuine prospective decision system. From a professional perspective, it would be misleading to present such a result as though it directly implied deployment readiness.

6.2 Medical-style diagnosis

The Breast Cancer task carries the clearest safety implications. In a medical context, false negatives are especially serious because a malignant case may be missed or delayed. That means strong accuracy alone is not enough. Professional evaluation must place particular weight on recall, F_1, and ranking-based measures, because these better reflect the cost of missed malignant cases.

At the same time, even strong benchmark performance should be treated cautiously. The project did not use a clinical validation pipeline, did not assess hospital workflow integration, and did not examine subgroup effects or institutional variation. The results therefore have academic value as a classification benchmark, but should not be described as clinically deployable.

A further issue is data realism. In the authoritative Breast Cancer pipeline, the identifier column was retained. Although this does not automatically invalidate the benchmark, it reduces the realism of the feature representation and should be acknowledged openly. In high-stakes domains, honest reporting of such caveats is part of professional responsibility.

6.3 Cybersecurity and Bitcoin Heist

The Bitcoin Heist case raises the most important professional issues in the project. In a cybersecurity setting, false negatives may allow harmful activity to remain undetected, while false positives may create analyst overload, waste investigative resources, and unfairly implicate benign actors. This makes threshold choice and score interpretation professionally important rather than merely technical.

The strongest professional lesson from this dataset is the danger of inflated claims caused by weak evaluation. Earlier random-split Bitcoin experiments produced unrealistically strong results. The stricter report-era pipeline, which removed problematic fields, used grouped cross-validation by year, and evaluated on an unseen temporal holdout, produced far weaker but much more believable performance. This is not a weakness of the project. On the contrary, it is one of its most professionally valuable outcomes.

In security applications especially, overclaiming performance can create a false sense of assurance. A benchmark that appears excellent under unrealistic evaluation may encourage poor downstream decisions, inappropriate trust, or wasteful operational deployment. The responsible professional response is therefore not to preserve flattering numbers, but to challenge them when they appear suspicious and report the corrected picture honestly.

7. Reproducibility, transparency, and honest reporting

A major professional responsibility in machine learning is to report results honestly and reproducibly. This includes documenting preprocessing, evaluation design, hyperparameter selection, limitations, and the conditions under which claims are valid. It also includes being explicit when some evidence is stronger than other evidence.

This project makes a positive contribution here through its unified experimentation framework, saved outputs, matched baseline comparisons, and explicit treatment of threats to validity. Most importantly, it does not conceal the fact that some parts of the benchmark are more fragile than others. The retention of the Breast Cancer identifier field, the presence of the Bank Marketing `duration` feature, and the sharp Bitcoin performance drop under stricter evaluation are all acknowledged directly.

This matters professionally because reproducibility is not just a convenience. It is part of engineering and scientific integrity. If a result cannot be traced back to a clear pipeline, or if strong claims depend on fragile evaluation choices, then confidence in the work should be reduced accordingly.

8. Professional judgement and summary

The professional issues in this project are not separate from the technical work; they are part of it. The same design decisions that affect metrics also affect fairness, privacy, safety, operational burden, and the honesty of the conclusions that can be drawn.

The strongest professional stance supported by the project is therefore a cautious one. The custom models are academically useful and, on several datasets, competitively implemented. However, none of the case studies justifies a simple claim of deployment readiness. Instead, the project supports a more mature conclusion: trustworthy evaluation, transparent reporting, explicit attribution, and clear recognition of limitations are as important as model selection itself.

That conclusion is particularly relevant to the banking, medical, and cybersecurity examples used here. In each domain, the cost of overclaiming can exceed the value of a superficially impressive benchmark. A professionally responsible machine learning project must therefore do more than optimise scores. It must state what the system can genuinely support, what it cannot yet support, and what further safeguards would be required before any real-world use could be justified.

References

References

[1] Association for Computing Machinery. *ACM Code of Ethics and Professional Conduct*. ACM, 2018.

[2] British Computer Society. *BCS Code of Conduct*. BCS, The Chartered Institute for IT.

[3] Information Commissioner's Office. *Guidance on AI and data protection*. ICO, 2023.

[4] National Cyber Security Centre. *Guidelines for secure AI system development*. NCSC, 2023.

[5] S. Barocas, M. Hardt, and A. Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, Cambridge, MA, 2023.

[6] S. Moro, P. Cortez, and P. Rita. A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62:22–31, 2014.

[7] C. G. Akcora, Y. Li, Y. R. Gel, and M. Kantarcioglu. BitcoinHeist: Topological data analysis for ransomware prediction on the Bitcoin blockchain. In *Proceedings of the*

Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20), pages 4439–4445, 2020.